

tel. USB 2.0 is in wide use today, while USB 3.0 is catching up, with consumer products using it having been introduced last year. Wireless USB adds a wire-free dimension to this universal standard using ultra-wideband wireless technologies.

USB is also quite fast. With the introduction of the third generation of this standard, it has overtaken the speed of even eSATA. Theoretically, the maximum data rate of USB 2.0 and Wireless USB is 480 Mbps and USB 3.0 can operate at 5 Gbps. USB is now used not only in computers but also in smart phones and other handhelds, not just for communication and data transfer but also as a power chord. However, Intel has something more in store.

In July 2010, Intel Corporation demonstrated a way of using light beams instead of electrons to carry data. Codenamed Light Peak, this research prototype comprises the world's first silicon-based optical data connection with integrated lasers. With speeds of up to 50 Gbps, not only is the link much faster than today's copper-wire based technologies, it is also capable of spanning longer distances. The first generation itself is expected to operate at around 10 Gbps, which is two times faster than USB 3.0. While this speed itself is sufficient to transmit a full high-definition (HD) movie in a matter of seconds, it is believed that Light Peak will attain a speed of 100 Gbps within a decade.

"While telecommunications and other applications already use lasers to transmit information, current technologies may not be apt for PC applications," says Gupta.

Another popular general-purpose I/O standard is the peripheral component interconnect express (PCIe), now in generation 2.0, with 3.0 expected anytime now. "PCIe 3.0 is the next emerging high-speed interface technology that will impact the PC and workstation platforms," says Dayakar Reddy, managing director, MosChip Semiconductor.

Testing high-speed serial interfaces

Driven by consumer demand, design engineers constantly need to integrate new functionality into their designs, incorporating cutting-edge technology. To aid the engineers, the design and operation of each of the serial data technologies is defined in a unique standards document—usually the product of an industry committee. Each standard also requires specialised measurements and compliance testing procedures, which add engineering work beyond the already complex design task. The standards address data signalling and encoding, packetisation, clock embedding, transmission properties and compliance test procedures. With clear standards and test procedures for high-speed serial technologies, designers are able to create products that provide off-the-shelf interoperability.

Typically, high-speed serial data test phases include:

- **Design verification.** Design and test engineers need to ensure that the real-world operation of their designs meets the design simulation and technical performance specifications, and that they are fully functional. This requires complete characterisation, debug and analysis under realistic and limit stress conditions.
- **Characterisation.** During the initial testing of a component, designers characterise performance to determine whether the signal behaviours are within specification. A host of precise measurements need to be made, often in real time, to determine whether the high-speed buses are working properly. Characterisation typically includes measuring rise/fall times, edge-to-edge timing, jitter and jitter tolerances, signal path skew, bus settling times, and data path variations.
- **Troubleshooting.** Any discovered characterisation or operational issues require engineers to debug and isolate failures or anomaly conditions such as hardware timing, crosstalk, signal quality and/or software design problems.
- **Compliance test.** Ensuring plug-n-play compatibility between multiple vendors and products requires confirmation of final designs to industry-specific serial data standards. Compliance testing can be a complex and potentially time-consuming step to ensure interoperability and usually requires advanced testing and analysis capabilities. Serial standards normally include extensive amplitude, timing, jitter, impedance and eye diagram measurements within their compliance testing specifications.

The whole process requires a range of test equipment and solutions including real-time oscilloscopes, automation tools to support transceiver testing, multi-level transmission testing and receiver margin testing, network analysis and physical layer testing, TDR-based network analysis, arbitrary waveform generator for transmitter/receiver testing, differential probing, protocol analyser module solutions and more!

—Evan Sun, technical marketing manager, Tektronix-Asia Pacific Region

The PCIe specification is a high-performance, general-purpose I/O interconnect defined for a wide variety of computing and communication platforms. It can be used as an interconnect technology at the motherboard level or as a computer expansion slot for add-in boards. This serial technology is replacing legacy parallel PCI/PCI-X buses used across PC and server applications. While 1.0 supports a single-lane data rate of 2.5 GT/s, generation 2.0 doubles that rate to 5 GT/s with 8-/10-bit encoding. The specifications of the upcoming version 3.0 provide for an 8 GT/s link that achieves double the data rate of 2.0 with data scrambling.

Speaking of expansion slots, the Intel-pioneered unified extensible firmware interface or UEFI (specification published in 2009) is meant to enable faster boot times and pave an easier path for new hardware. The extensible nature of this new interface helps ensure that it will be capable of dealing with whatever new PC expansions hardware makers can dream up in the next three decades.

It is a networked world, really

This is truly the age of the network. Almost every device is networkable, from test equipment to machines in a factory. As a result, networking